

Sri Lanka Institute of Information Technology



# **HostelMS — Hostel Management Mobile Application**

## **WD-IT-33**

SE2020 – Web and Mobile Technologies

Group Assignment  
Year 2 Semester 2 (2026)

**GitHub Repository:**

**<https://github.com/Kalindu-Pabasara/Hostel-Management-Mobile.git>**

# Problem Statement

## 1. Context

Managing a student hostel involves complex operations including room allocations, maintenance tracking, visitor logs, and financial management. Many educational institutions still rely on manual, paper-based systems or fragmented digital spreadsheets to handle these tasks.

## 2. Problem Description

The current manual processes face several critical issues:

- **Inefficient Allocation:** Tracking room occupancy across multiple floors and room types (Single, Double, Suite) is prone to errors, leading to overbooking or underutilization.
- **Maintenance Delays:** Students lack a centralized platform to report issues (leaks, electrical faults). Requests are often lost or forgotten, leading to student dissatisfaction.
- **Security Gaps:** Visitor tracking is often handled via physical logbooks that are difficult to search and maintain for security audits.
- **Financial Complexity:** Managing recurring fees and payment proofs manually is time-consuming and lacks transparency for students.
- **Information Silos:** Data is not shared in real-time between students and administrators, causing communication breakdowns.

## 3. Proposed Solution

**HostelMS** is a Full Stack Mobile Application designed to centralize and automate hostel operations.

- **Real-time Dashboard:** Providing administrators with instant stats on occupancy and revenue.
- **Automated Workflow:** Enabling students to book rooms, log maintenance requests, and upload payment proofs directly from their mobile devices.
- **Centralized Security:** Digital visitor registration linked to student hosts for accountability.
- **Role-Based Access:** Ensuring data security by separating Administrative controls from Student self-service features.

#### **4. Objectives**

- To digitize 100% of room allocation and maintenance records.
- To reduce maintenance response time through real-time notifications.
- To provide a premium, modern UI/UX for students to manage their hostel life.